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(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU & RTMNU



B. Tech Summer / Winter - 2025
(ESE / Make-up / Supplementary) Examination

Branch: Computer Science & Engineering

Sem:- V

Subject: Machine Learning

Subject Code: CS5T002

Marks: 100

Date of Exam: 30/10/25 Time: 3 Hrs.

Instructions to the students:

1. Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.
2. All the questions carry marks as indicated.
3. Assume suitable data whenever necessary.
4. Illustrate your answer whenever necessary with the help of neat sketches.
5. Use of Programmable Calculator is prohibited.
6. Don't use red pen for writing the answers.

COs: Students should be able to:

1. Understand the basics of machine learning and different types of learning methods.
2. Apply concept learning and clustering algorithms to solve learning problems.
3. Analyze Bayesian learning approaches and learning theories.
4. Design predictive models using regression, kernel methods, and ensemble techniques.
5. Evaluate and compare machine learning algorithms using decision trees and hypothesis testing.

SECTION A

1. All Questions are compulsory.
2. Answer in one/two word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define Machine Learning.	L1/CO-1	2
Q.2	State different Steps involved in the K-mean Algorithm.	L1/CO-2	2
Q.3	Identify the Concept of Hierarchical Clustering.	L2/CO-2	2
Q.4	List out the Different Issues in Machine Learning	L1/CO-1	2
Q.5	Define Hypothesis Space.	L2/CO-5	2
Q.6	Tell about the Well Posed Learning Problems with Examples.	L2/CO-1	2
Q.7	Restate the Decision Tree Algorithm.	L2/CO-5	2
Q.8	Write a Short note on Reinforcement Learning.	L3/CO-1	2
Q.9	Give the Advantages and Disadvantages of Supervised Learning.	L2/CO-1	2
Q.10	Define the following terminology w.r.t. Decision Tree. a) Root Node. b) Leaf Node. c) Splitting. d) Pruning	L1/CO-5	2
Q.11	State the Applications of Clustering.	L1/CO-2	2

Q.12 From the deck of Cards, Find the Probability of the card being picked is King Given that it is the Face Card.

L1/CO-3

2

- This can be represented as: $p(\text{King} | \text{Face})$.
- There are total 52 cards in a deck ($n=52$), from Which 12 cards are Face Cards: King, Queen and Jack with Club, Diamond, heart and Spade.
- There is a set of Face cards with 12 members.
- There is a Subset of King Cards with 4-members.
- So, 4 Face cards out of 12 Face cards ($4/12$) are King cards. Use Bayes Theorem to Calculate.

SECTION B

1. All Questions are compulsory.

Q. No.	Questions	Level/CO Mapped	Marks
Q.13	Distinguish between Supervised and Unsupervised Learning.	L4/CO-1	4
Q.14	Illustrate the concept of the Bayes Theorem and illustrate it using its formula.	L3/CO-3	4
Q.15	Discuss the Hypothesis Space Search in Decision Tree in detail.	L2/CO-5	4
Q.16	Examine Semi-supervised Learning With Example and Advantages and Disadvantages.	L4/CO-1	4
Q.17	Write short note on K-mean Clustering.	L3/CO-2	4
Q.18	For the Given Dataset, Apply Naïve –Bayes Algorithm and Predict the Outcome for a car, Car= {Red, Domestic, SUV}	L3/CO-3	4

Color	Type	Origin	Stolen
Red	Sports	Domestic	Yes
Red	Sports	Domestic	No
Red	Sports	Domestic	Yes
Yellow	Sports	Domestic	No
Yellow	Sports	Imported	Yes
Yellow	SUV	Imported	No
Yellow	SUV	Imported	Yes
Yellow	SUV	Domestic	No
Red	SUV	Imported	No
Red	Sports	Imported	Yes

SECTION C

1. Solve any FOUR Questions.

Q. No.	Questions	Level/CO Mapped	Marks
Q.19	Illustrate in Detail Concept of Decision Tree and also Outline the main steps followed in constructing a Decision Tree algorithm.	L4/CO-5	7
Q.20	Explain the Concept of Hierarchical Agglomerative Clustering (HAC) and illustrate its workflow with a diagram.	L5/CO-2	7
Q.21	Analyse the Following Concept and illustrate it using its formula. a) Naïve Bayes Classifier. b) Bayes Optimal Classifier	L4/CO-3	7
Q.22	Examine in detail about Hypothesis Testing with their Types.	L4/CO-5	7
Q.23	Describe Regression in detail and also describe Linear Regression with their Types.	L5/CO-4	7

- Q.24 Given the Training data in the table (Buy Computer Data), Predict the Class of the Following New Example Using Bayes Optimal Classifier.
Buy Computer = {Age = ≤ 30 , income = medium, student = yes, Credit-Rating = fair}

L5/CO-3

7

Age	Income	Student	Credit Rating	Buy's Computer
≤ 30	High	No	Fair	No
≤ 30	High	No	Excellent	No
31...40	High	No	Fair	Yes
≥ 40	Medium	No	Fair	Yes
≥ 40	Low	Yes	Fair	Yes
≥ 40	Low	Yes	Excellent	No
31...40	Low	Yes	Excellent	Yes
≤ 30	Medium	No	Fair	No
≤ 30	Low	Yes	Fair	Yes
≥ 40	Medium	Yes	Fair	Yes
≤ 30	Medium	Yes	Excellent	Yes
31..40	Medium	No	Excellent	Yes
31..40	High	Yes	Fair	Yes
≥ 40	Medium	No	Excellent	No

SECTION D

1. Solve any TWO Questions.

Q. No. Questions

Level/CO
Mapped

Marks

- Q.25 Rewrite the Concept of Basic Sampling Theory and also Rewrite following about Sampling:

L5/CO-5

12

- Process of Sampling
- Types of Sampling

- Q.26 Consider the Given Dataset, Apply Naïve Baye's Algorithm and Predict that if a Tennis Play has following Properties

L6/CO-3

12

Tennis Play = {Outlook=Sunny, Humidity=High, Wind= Strong}
Tennis Play = {Outlook=Sunny, Humidity=High, Wind= Weak}

Day	Outlook	Humidity	Wind	Play Tennis
D1	Sunny	High	Weak	No
D2	Sunny	High	Strong	No
D3	Overcast	High	Weak	Yes
D4	Rain	High	Weak	Yes
D5	Rain	Normal	Weak	Yes
D6	Rain	Normal	Strong	No
D7	Overcast	Normal	Strong	Yes
D8	Sunny	High	Weak	No
D9	Sunny	Normal	Weak	Yes
D10	Rain	Normal	Weak	Yes
D11	Sunny	Normal	Strong	Yes
D12	Overcast	High	Strong	Yes
D13	Overcast	Normal	Weak	Yes
D14	Rain	High	Strong	No

- Q.27 Describe the Following About Machine Learning:

L6/CO-1

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- Applications of Machine Learning
- Life Cycle of Machine Learning